Unit 4 · Outcome 2 · Energy Sources

A Sustainable Energy Future

KK10 VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.

The big idea: a future that runs on less

1 The big idea: a future that runs on less orientation Every earlier dot point in this chapter asked "where does our energy come from?" . This one asks the harder, more examinable question: how do we build an energy system that can keep going indefinitely without wrecking the systems we depend on? The study design gives you three levers to pull — and a good answer pulls all three, not just one. The leaky-bucket analogy Picture our energy demand as a bucket you're trying to keep full. The obvious move is to pour in more water — build more power stations

Lever 2 — Replacing fossil fuels (interactive)

2 Lever 2 — Replacing fossil fuels (interactive) 3D · replace The Kalduke Valley's Wirndook brown-coal station (our chapter's villain from KK03) is closing. In its place, the Mirran Plains Renewable Energy Zone rises — wind, solar, big batteries and a new transmission line. Drag the year slider to watch the grid transform, and watch what happens to renewables share and emissions. Scene A Kalduke Valley grid · 2025 → 2045 drag the year · orbit with mouse 3D view needs WebGL — the readouts and slider below still work fully. Year 2025 Renewables share 42 %

What you replace	With what	The catch
Coal & gas generation	Wind, solar, hydro, plus a controlling-stake public builder (Victoria's revived SEC , HQ in Morwell)	Output is intermittent — it must be firmed
The gap when the sun sets	Storage : pumped hydro (Snowy 2.0), grid batteries (VIC targets 2.6 GW by 2030, 6.3 GW by 2035)	Storage is finite-duration and costs money & materials
Petrol & gas you burn at home	Electrification : EVs, heat pumps, induction — shift the load onto a clean grid	Only as clean as the grid charging it
The wires themselves	New transmission : VNI West, Western Renewables Link, Marinus Link	Build delays & social licence are the real bottleneck

Lever 1 — Efficiency: resource vs conversion

3 Lever 1 — Efficiency: resource vs conversion concept The dot point names two distinct kinds of efficiency. Examiners love to test whether you can tell them apart, so lock this distinction down. Resource efficiency Less raw resource per service Using fewer units of the primary resource to deliver the same benefit. Example: a cogeneration plant that captures "waste" heat to warm buildings extracts more useful service from each tonne of gas. Recycling aluminium uses ~5% of the energy of smelting new metal — same product, far less resource. Conversion e

Conversion efficiency, made visible (interactive)

4 Conversion efficiency, made visible (interactive) 3D · efficiency Pick a head-to-head and watch where the energy actually goes. The flowing particles show 100 units of input splitting into useful output and wasted heat . The better technology keeps more particles on the useful path. Scene B Where does the energy go? choose a comparison 3D view needs WebGL — the efficiency figures below still work fully. Lighting Heating Transport Incandescent globe 5 % useful LED globe 40 % useful Energy saved, same service 88 % Why transport is the headline wi

Lever 3 — Reducing personal consumption (interactive)

5 Lever 3 — Reducing personal consumption (interactive) 3D · reduce This is the lever you control. Toggle each measure on the Kalduke Valley demonstration home and watch its yearly energy use, power bill and emissions fall. Notice how cheap behaviour changes stack up against big-ticket upgrades. Scene C The negawatt house toggle the measures 3D view needs WebGL — the toggles and readouts below still work fully. Insulate & draught-proof LED lighting Heat pump (reverse-cycle) Behaviour (1°C, shorter showers) Rooftop solar + battery Switch pe

Putting it together: evaluating the options

6 Putting it together: evaluating the options synthesis A top-band answer doesn't treat the three levers as a menu to pick from — it shows how they interlock and judges each against the sustainability principles (KK07) and base/peak load reliability (KK09). Option Advantage Disadvantage / limit Sustainability link Improve efficiency Cheapest & fastest; cuts demand, emissions and bills at once Bounded by the 2nd Law — you can never reach 100%; rebound effect can erode gains Intergenerational equity; conservation of resources Replace fossil fuels Tackles t

Option	Advantage	Disadvantage / limit	Sustainability link
Improve efficiency	Cheapest & fastest; cuts demand, emissions and bills at once	Bounded by the 2nd Law — you can never reach 100%; rebound effect can erode gains	Intergenerational equity; conservation of resources
Replace fossil fuels	Tackles the root cause — finite stocks & CO_2; renewables are now lowest-cost new supply	Intermittent → needs storage & transmission; material & land footprint; transition jobs	Precautionary principle; intergenerational equity
Reduce consumption	Negawatts need no new infrastructure; empowers individuals; eases peak load	Relies on behaviour & equity of access; hard to mandate; limited ceiling alone	Intergenerational & intragenerational equity

Command-word decoder

◈ **COMMAND WORDS**

7 Command-word decoder exam skill Same content, different verb, different answer. Tap each card to see exactly what the examiner wants — and how many marks it's usually worth. Identify / State Name a relevant option. No explanation needed. One short phrase. ~1 mark Describe Say what the option is and how it works — features and characteristics, no judgement. 2–3 marks Explain Give the how/why — link cause to effect (e.g. why a heat pump cuts energy use). 2–4 marks Compare Show similarities and differences between options, point by point. Use "whereas". 3

Six exam traps that cost marks

⚠ **EXAM TRAPS**

8 Six exam traps that cost marks pitfalls These are the recurring ways students lose marks on this dot point. Each is a trap many bright students walk into. ▶ Confusing "renewable" with "sustainable" trap 1 A source can be renewable but used unsustainably (e.g. clearing native forest for biomass), or non-renewable yet relatively low-emission in transition (gas as a "firming" fuel). Sustainability is about the whole system and the principles (KK07) — not a label on the fuel. Don't equate the two. ▶ Only talking about generation trap 2 Replacing coal needs

P·E·L writing trainer

✎ P·E·L WRITING

9 P·E·L writing trainer write & self-check Strong extended responses follow P·E·L : make a Point , Explain the mechanism, then Link back to the question (sustainability, base/peak, or the judgement). Write your answer below — the trainer lights up the key ideas as you include them. "Evaluate improving energy efficiency as a strategy for a sustainable energy future." (4 marks) Aim for a Point, an Explanation of the mechanism, and a Link to a judgement / sustainability principle. 0 / 6 key ideas Clear Now compare a model good and bad answer to the same que

P · POINT answer the verb → **E · EVIDENCE** example / mechanism → **L · LINK** tie back to question

The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

VCAA-style practice · 20 marks

★ PRACTICE & SELF-MARK

10 VCAA-style practice · 20 marks self-mark Five questions building from recall to full evaluation. Attempt each in your logbook first, then reveal the model answer and marking scheme and award yourself stars. Your progress saves automatically. Q1 2 marks Distinguish between resource efficiency and conversion efficiency , giving one example of each. Command word: Distinguish Reveal model answer & scheme Model answer Resource efficiency means using fewer units of the primary resource to deliver the same service (e.g. recycling aluminium uses ~5% of the en

One-screen cheat sheet

✦ RECAP / CHEAT SHEET

11 One-screen cheat sheet revision KK10 — A sustainable energy future Improve efficiency Resource : less raw fuel per service (cogeneration, recycling) Conversion : less waste per step (LED, CCGT, heat pump COP 3–4) Cheapest, fastest lever = "negawatts" Limit: 2nd Law + rebound effect Replace fossil fuels Renewables + storage + transmission + electrification VIC: 95% renewables 2035, net-zero 2045 Coal out: Yallourn '28, Loy Yang B '32, A '35 Limit: intermittency → must be firmed Reduce consumption Insulate, draught-proof, star-rated appliances

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.

P · POINT answer the verb → **E · EVIDENCE** example / mechanism → **L · LINK** tie back to question

The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

Incandescent globe 5 % useful

✓ STRONG / MODEL ANSWER

HIGH-BAND RESPONSE Improving efficiency reduces the energy needed for the same service , e.g. a heat pump delivers 3–4 units of heat per unit of electricity (COP 3–4). This cuts demand, emissions and cost simultaneously , and is the fastest, cheapest lever because it needs no new generation — every avoided unit is a "negawatt". However it is bounded by the Second Law and the rebound effect , so it cannot decarbonise alone. Efficiency is therefore highly suitable as a first step that shrinks the system we must then replace and firm. Point + mechanism + advantage + limitation + justified judgement = top band.

✗ WEAK ANSWER

LOW-BAND RESPONSE Efficiency is good because it saves energy and helps the environment. We should use LED lights and turn things off. This is renewable and sustainable so it is a good idea for the future. Everyone should try to be efficient to stop climate change. No mechanism, no figures, no limitation, no judgement; "renewable" misused; vague. Caps low.

✓ STRONG / MODEL ANSWER

Resource efficiency means using fewer units of the primary resource to deliver the same service (e.g. recycling aluminium uses ~5% of the energy of smelting new metal). Conversion efficiency is the fraction of input energy that becomes useful output at a transformation step (e.g. a combined-cycle gas turbine ~55–60% vs open-cycle ~35%). Marking scheme 1 mark — correct distinction (less resource used vs less waste per conversion). 1 mark — one valid example for each type.

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.